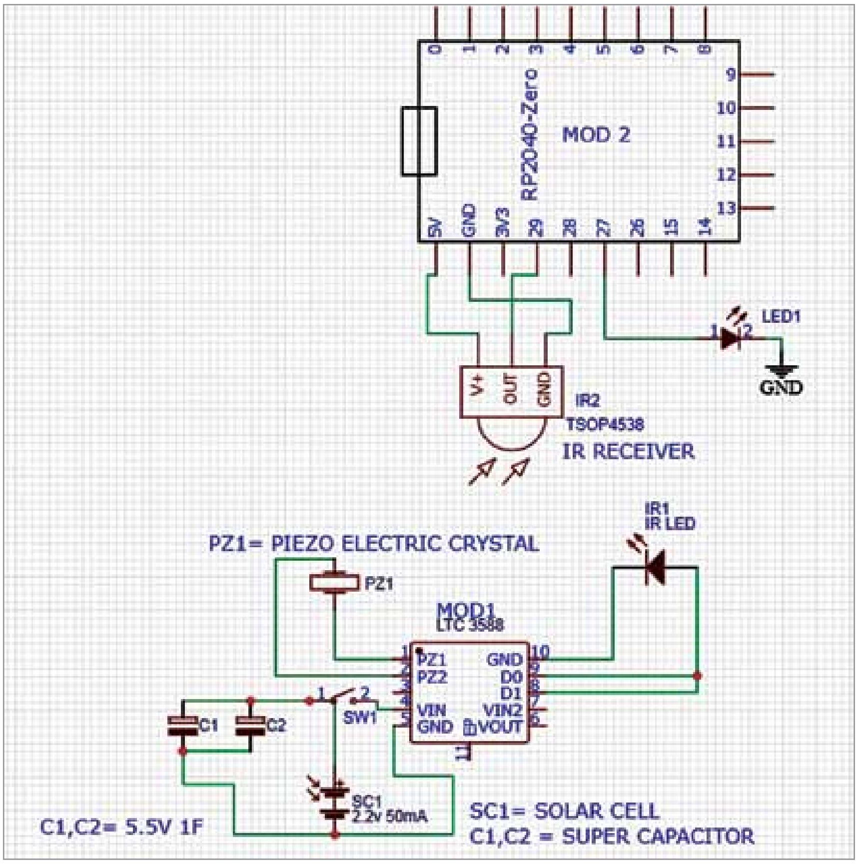

Fig. 3: The receiver section, including transmitter

```
void setup() {
  // initialize the LED pin as an output:
  pinMode(29, 0);
  // initialize the pushbutton pin as an input:
  pinMode(27, 1);
}
void loop() {
  if (analogRead(29)>= 10) {//// you can cahnage the
  //funcrtion or even read the hex value and set
  //the function to be perform when the signal is recived
    digitalWrite(27, HIGH);
  } else {
    digitalWrite(29, LOW);
  }
}
```

Fig. 4: Code for the receiver

(LED1). When the transmitter send the signal through IR1, it is received by IR2 in the receiver. It is necessary to ensure aligment of LED1 and LED2.

The V+, OUT, and GND pins of TSOP4538 are connected to the 5V supply, pin 29, and GND pins of MOD2, respectively. The program in source code, which is shown in Fig. 4, controls the LED. Any other function can also be included in the program for the IR signal received from the IR remote.

The components needed for this project are listed under the Bill of Material table.

The beauty of this project is that the kinetic energy used for pressing a key on the remote is harvested and used to charge the capacitors. This mechanical energy is converted to electric energy by the circuit. A small solar cell is also used to harvest photonic energy from the ambient light in the room.

All of the energy harvested is stored in the super capacitors and the remote uses this stored energy to work. Thus, whenever a switch on the remote is pressed, the electric energy is used to light the IR LED and a signal is given to the receiver to perform the desired function.

Software

The operation of remote receiver circuit can be tested by using the software program, which is written in Arduino programming language Sketch and saved as batteryless.ino. The latest Arduino IDE should be used to compile and upload the program.

During uploading, you need to select the the RP2040 as board in Arduino IDE for compiling and uploading the code shown in Fig. 4.

Construction and testing

After uploading the source code batteryless.ino into MOD1 and assembling the complete circuit of transmitter and recceiver, align the circuit in such a way that IR1 light falls on IR2 (refer Fig. 3) when the piezoelectric crystal is pressed. Please ensure that the solar cell is also connected to the super capacitors to keep them charged.

Working of the circuit is simple. When the piezoelectric crystal is pressed, the generated energy makes the IR LED (IR1) glow, thus sending a signal to the receiver circuit. This signal is received by IR2 in the receiver to perform the function as per the program. Thus, the remote can work for years without any battery.

The remote can be used even in the dark, with a slight press on the piezo element. EFY

EFY Note
The source code of this project is available for free download at *source.efymag.com*

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